Job-Skill Matching with Skill Type Classification

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Abstract

This paper addresses the task of Job-Skill Matching with Skill Type Classification. The objective is to retrieve relevant professional skills for a given job title and further classify each skill into one of three categories: core, complementary, or transversal. The task combines information retrieval and multi-label classification, requiring models that understand semantic relationships between job titles and skills while distinguishing essential competencies from contextual or broad transferable skills. We review prior work on skill extraction, occupational classification, domain-specific embeddings, and transversal skill detection to inform our methodological approach.

Keywords

Job-Skill Matching, Skill Classification, Information Retrieval, Transformers, Multi-label Classification

1. Introduction To The Task

The Job-Skill Matching with Skill Type Classification task focuses on two interconnected objectives. First, given a job title, the system must retrieve the most relevant professional skills associated with that role. Second, each retrieved skill must be classified into one of three categories: core, complementary, or transversal.

Core skills refer to competencies that are essential for performing a job independently of the organization or context. Complementary skills support the role but may vary depending on industry or employer requirements. Transversal skills represent broad, transferable abilities such as communication or teamwork that apply across multiple occupations.

The dataset consists of three components. The training set includes more than 5,000 job titles paired with professional skills. The development set contains 200 job titles with skills normalized to ESCO terminology, along with annotations specifying whether each skill is core or contextual. The test set includes 500 job titles for which participants must predict both relevant skills and their corresponding skill types using a provided gazetteer.

Model performance will be evaluated using information retrieval metrics such as Normalized Discounted Cumulative Gain (nDCG) and Mean Average Precision (MAP), along with Mean Reciprocal Rank (MRR) and Precision@K. This evaluation framework emphasizes both ranking quality and classification accuracy.

2. Literature Review

Research on job-skill modeling has developed along three major directions: skill extraction from job advertisements, occupational classification, and skill categorization. Since our task requires both retrieval and classification, insights from each of these areas are relevant.

Recent work on job description parsing proposes an explainable transformer-based ensemble model designed to extract both technical and non-technical skills from job postings [1]. The model combines convolutional layers to capture local phrase structures, bidirectional LSTMs to model long-range dependencies, and a Conditional Random Field (CRF) layer to refine sequence labeling decisions. These components operate on contextual embeddings generated by BERT. Using a manually annotated dataset of IT job roles, the system reports strong performance compared to classical baselines and also applies LIME to interpret model predictions. The authors further note challenges such as limited annotated

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data and difficulty recognizing newly emerging skills, which is relevant when predicting unseen skills in our task.

Another relevant line of research addresses hierarchical occupational classification [2]. A transformer-based system fine-tuned on job posting text demonstrates how occupations can be mapped to structured taxonomies such as CIUO-08/ISCO-08. The study highlights the risk of hierarchical error propagation, where early mistakes cause larger downstream errors. While our task does not use a hierarchy of occupational labels, we similarly require careful differentiation between closely related skill types, where small errors can affect ranking-based evaluation.

Skill categorization has also been explored through domain-specific language modeling. SkillBERT introduces a BERT model trained from scratch on a large corpus of job requisitions to better represent skills and their relationships [3]. The study derives embedding-based features and applies downstream classifiers for multi-label categorization, showing that domain-trained representations can outperform generic embeddings. This is relevant to our setting because skill meaning often depends on labor-market context rather than general language usage.

Transversal skill detection presents an additional layer of complexity. Transversal skills are often implied rather than explicitly listed, making them harder to identify reliably [4]. The study proposes specialized neural classifiers and applies augmentation and sampling to reduce imbalance. It also introduces keyword extraction methods for interpretability, which is useful when justifying why a certain skill is categorized as transversal rather than core or complementary.

Across these studies, several findings are consistent: transformer-based contextual embeddings outperform traditional approaches, domain adaptation improves representation quality, and interpretability is valuable for validating decisions. Our task builds on these insights by combining retrieval and skill-type classification under ranking-based evaluation metrics.

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